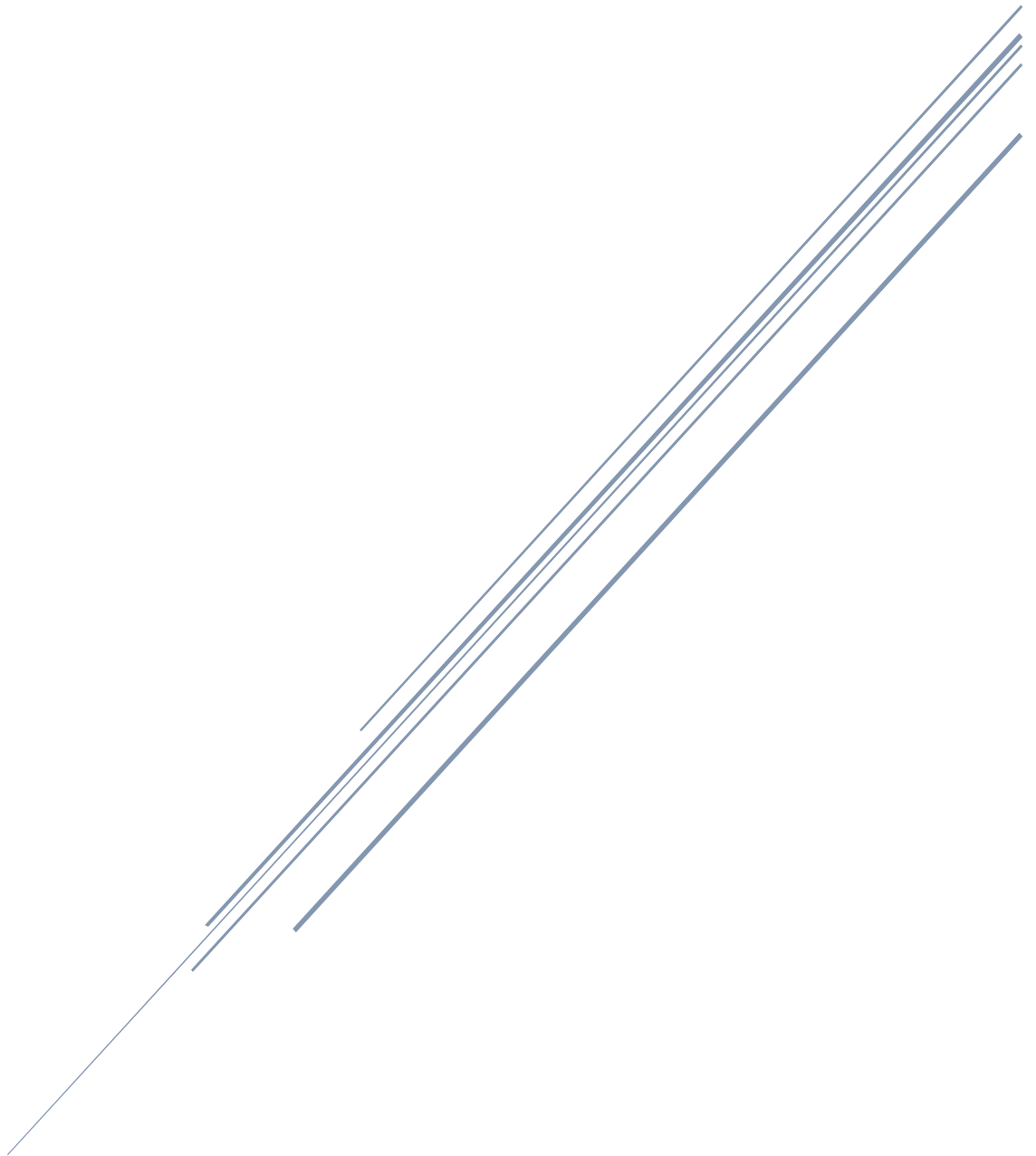


CHAPTER 3

Linear regression



3.1 Definition

Linear regression lives up to its name: it is a very straightforward approach for predicting a quantitative response Y on the basis of a predictor variable X .

We assume that the true relationship between X and Y takes the form $Y = f(X) + \epsilon$ for some unknown function f , where ϵ is a mean-zero random error term. If f is to be approximated by a linear function, then we can write this relationship as

$$Y = \beta_0 + \beta_1 X + \epsilon.$$

Here β_0 is the intercept term—that is, the expected value of Y when $X = 0$, and β_1 is the slope—the average increase in Y associated with a one-unit increase in X . The error term: the true relationship is probably not linear, there may be other variables that cause variation in Y , and there may be measurement error. We typically assume that the error term is independent of X . The model given by this equation defines the *population regression line*, which is the best linear approximation to the true relationship between X and Y .

3.2 Linear regression model

There are two very different algorithms or approaches for implementing the Linear Regression model: the “closed-form” solution approach known as the Ordinary Least Squares (OLS) method and the iterative optimization approach known as the Gradient Descent (GD).

3.2.1 Closed-form solution: Ordinary least squares (OLS)

OLS is a type of linear least squares method for estimating the unknown parameters in a linear regression model. OLS chooses the parameters of a linear function of a set of explanatory variables by the principle of least squares:

minimizing the sum of the squares of the differences between the observed dependent variable (values of the variable being predicted) in the given dataset and those predicted by the linear function.

OLS finds the optimal parameters by computing a closed-form solution for the **Normal equation**.

The OLS method is suitable for the following scenario.

- Dataset: not too large and fits into computer's memory
- No. of Features: not too large
- Relationship between input (features) and output (target): Linear

a) Simple linear regression

Example: Let's consider the simple linear regression model in which the price of a house is related to the number of square feet of living area (SQFT).

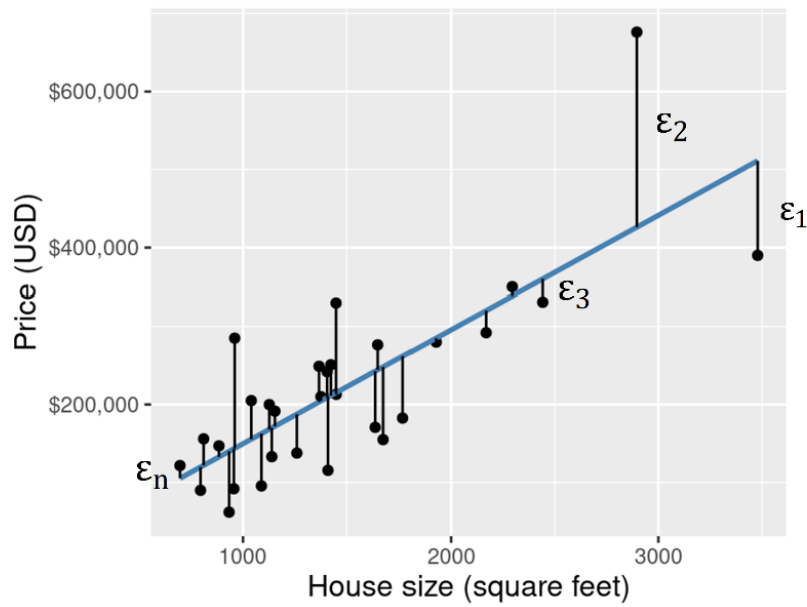


Figure 3.1 Scatter plot of sale price versus size

We have the system of equations:

$$\begin{cases} y_1 = \beta_0 + \beta_1 x_1 + \varepsilon_1 \\ y_2 = \beta_0 + \beta_1 x_2 + \varepsilon_2 \\ y_3 = \beta_0 + \beta_1 x_3 + \varepsilon_3 \\ \vdots \\ y_n = \beta_0 + \beta_1 x_n + \varepsilon_n \end{cases}$$

This system of equations can be written as:

$$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ 1 & x_3 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} + \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \vdots \\ \varepsilon_n \end{pmatrix}$$

So, we get

$$\varepsilon = Y - X\beta$$

The L2 norm of ε is defined by

$$\|\varepsilon\| = \sqrt{\varepsilon_1^2 + \varepsilon_2^2 + \varepsilon_3^2 + \dots + \varepsilon_n^2} = \|Y - X\beta\|$$

The objective is to minimize the sum of the squared errors

$$\min_{\beta_0, \beta_1} \|\varepsilon\|^2 = \min_{\beta_0, \beta_1} \|Y - X\beta\|^2 = \min_{\beta_0, \beta_1} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2 = \min_{\beta_0, \beta_1} E(\beta_0, \beta_1)$$

We have to find β_0 and β_1 such that:

$$\begin{cases} \frac{\partial E}{\partial \beta_0} = -2 \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) = 0 \\ \frac{\partial E}{\partial \beta_1} = -2 \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) x_i = 0 \end{cases}$$

$$\begin{cases} \sum_{i=1}^n y_i - n\beta_0 - \beta_1 \sum_{i=1}^n x_i = 0 \\ \sum_{i=1}^n y_i x_i - \beta_0 \sum_{i=1}^n x_i - \beta_1 \sum_{i=1}^n x_i^2 = 0 \end{cases}$$

$$\begin{cases} n\bar{y} - n\beta_0 - n\beta_1 \bar{x} = 0 \\ \sum_{i=1}^n y_i x_i - n\beta_0 \bar{x} - \beta_1 \sum_{i=1}^n x_i^2 = 0 \end{cases}$$

$$\begin{cases} \beta_0 = \bar{y} - \beta_1 \bar{x} \\ \sum_{i=1}^n y_i x_i - n\beta_0 \bar{x} - \beta_1 \sum_{i=1}^n x_i^2 = 0 \end{cases}$$

We plug the first equation in the second one, and we get

$$\sum_{i=1}^n y_i x_i - n\bar{x}(\bar{y} - \beta_1 \bar{x}) - \beta_1 \sum_{i=1}^n x_i^2 = 0$$

$$\sum_{i=1}^n y_i x_i - n\bar{x}\bar{y} + \beta_1 n\bar{x}^2 - \beta_1 \sum_{i=1}^n x_i^2 = 0$$

$$\sum_{i=1}^n y_i x_i - n\bar{x}\bar{y} + \beta_1 (n\bar{x}^2 - \sum_{i=1}^n x_i^2) = 0$$

$$\beta_1 = \frac{\sum_{i=1}^n y_i x_i - n\bar{x}\bar{y}}{\sum_{i=1}^n x_i^2 - n\bar{x}^2}$$

b) Multiple linear regression model

Example: Suppose that Y is the fuel consumption of a particular model of car in m.p.g.

Suppose that the attributes are:

1. X1 — the weight of the car.
2. X2 — the horse power.
3. X3 — the no. of cylinders.

Typically, the data will be available in the form of an array like this:

$$\begin{array}{cccc} y_1 & x_{11} & x_{12} & x_{13} \\ y_2 & x_{21} & x_{22} & x_{23} \\ \dots & & & \\ y_n & x_{n1} & x_{n2} & x_{n3} \end{array}$$

where n is the number of observations or cases in the dataset.

One very general form for the model would be

$$Y = f(X_1, X_2, X_3) + \epsilon$$

ϵ where f is some unknown function and ϵ is the error in this representation which is additive in this instance. Since we usually don't have enough data to try to estimate f directly, we usually have to assume that it has some more restricted form, perhaps linear as

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \varepsilon$$

Matrix Representation

Given the actual data, we may write

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \varepsilon_i \quad i = 1, \dots, n$$

But the use of subscripts becomes inconvenient and conceptually obscure. We will find it simpler both notationally and theoretically to use a matrix/vector representation. The regression equation is write

$$y = X\beta + \varepsilon$$

where $y = (y_1, \dots, y_n)^T$, $\beta = (\beta_0, \dots, \beta_3)^T$ and $\varepsilon = (\varepsilon_1, \dots, \varepsilon_n)^T$

$$X = \begin{pmatrix} 1 & x_{11} & x_{12} & x_{13} \\ 1 & x_{21} & x_{22} & x_{23} \\ \dots & & & \\ 1 & x_{n1} & x_{n2} & x_{n3} \end{pmatrix}$$

The column of ones incorporates the intercept term.

Estimating β

The problem is to find β such that $X\beta$ is close to Y . We might define the best estimate of β as that which minimizes the sum of the squared errors, $\varepsilon^T \varepsilon$. That is to say that the least squares estimate of β , called $\hat{\beta}$ minimizes

$$\|\varepsilon\|_2^2 = \varepsilon^T \varepsilon = (y - X\beta)^T (y - X\beta)$$

By applying the transpose operations, we get

$$\begin{aligned} & (y^T - (X\beta)^T)(y - X\beta) \\ & y^T y - y^T X\beta - (X\beta)^T y + (X\beta)^T (X\beta) \\ & y^T y - y^T X\beta - \beta^T X^T y + \beta^T X^T X\beta \end{aligned}$$

$y^T X\beta$ is a scalar (a 1x1 matrix), so we apply the symmetry rule: $a^T b = b^T a$

$$\begin{aligned} & y^T y - (X\beta)^T y - \beta^T X^T y + \beta^T X^T X\beta \\ & y^T y - \beta^T X^T y - \beta^T X^T y + \beta^T X^T X\beta \\ & y^T y - 2\beta^T X^T y + \beta^T X^T X\beta \end{aligned}$$

Differentiating with respect to β and setting to zero, we find that $\hat{\beta}_{OLS}$ satisfies

$$\begin{aligned} -2X^T y + 2X^T X\hat{\beta}_{OLS} &= 0 \\ X^T X\hat{\beta}_{OLS} &= X^T y \end{aligned}$$

Provided $X^T X$ is invertible, the OLS solution is

$$\hat{\beta}_{OLS} = (X^T X)^{-1} X^T y$$

$X^T X$ is called a Gram matrix, and captures relationships between features.

If collinearity between features is high, then $X^T X$ is almost singular (i.e., its determinant is close to zero). Because the OLS estimates depend upon $(X^T X)^{-1}$, we would have problems in computing $\hat{\beta}_{OLS}$. In such cases, small changes to the elements of X can lead to large variations in $(X^T X)^{-1}$, resulting in inflated coefficients in $\hat{\beta}_{OLS}$. As a result, the OLS estimates become unstable and highly sensitive to small changes in the data.

The variance of the estimated coefficients in Ordinary Least Squares (OLS) is given by:

$$\text{Var}(\hat{\beta}) = \sigma^2 (X^T X)^{-1}$$

- This makes $(X^T X)^{-1}$ very large, which increases the variance of $\hat{\beta}$. Different samples from the data can produce very different estimates of $\hat{\beta}$. The least square estimator $\hat{\beta}$ may provide a good fit to the training data, but it will not fit sufficiently well to the test data.

3.2.2 Regression shrinkage (regularization) methods

3.2.2.1 Ridge Regression

One way out of this situation is to abandon the requirement of an unbiased estimator.

The objective is to minimize the sum of the squared errors

$$\min_{\beta} \|\varepsilon\|^2 = \min_{\beta} \|Y - X\beta\|^2, \text{ s. t. } \|\beta\|^2 \leq c^2$$

Geometric Interpretation of Ridge Regression

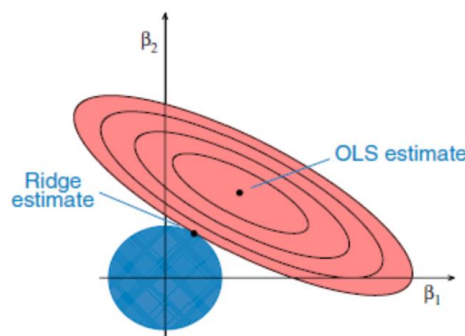


Figure 3.2 Estimation picture of ridge regression. Shown are contours of the error and constraint functions. The solid blue area is the constraint region $\beta_1^2 + \beta_2^2 \leq c^2$, while the red ellipses are the contours of the least squares error function. Ridge shrinks all coefficients but does not eliminate any; coefficients are smaller but nonzero.

The ellipses correspond to the contours of the sum of the squared errors: the inner ellipse has smaller sum of the squared errors, and the sum of the squared errors is minimized at ordinal least square (OLS) estimates.

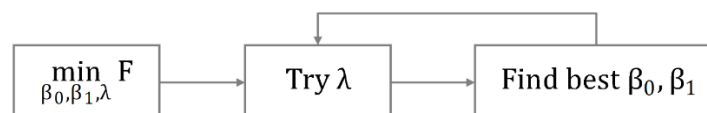
For $p=2$, the constraint in ridge regression corresponds to a circle, $\sum_{j=1}^p \beta_j^2 \leq c^2$

We want to minimize:

$$\min_{\beta} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2, \text{ s. t. } \beta_0^2 + \beta_1^2 \leq c^2$$

Instead of solving this as a constrained optimization problem, it is typically rewritten using a Lagrange multiplier (λ) and solved as an unconstrained optimization problem.

$$\min_{\beta} F(\beta_0, \beta_1, \lambda) = \min_{\beta} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2 + \lambda(\beta_0^2 + \beta_1^2 - c^2)$$



So, we have to minimize

$$\min_{\beta} \|Y - X\beta\|^2 + \lambda \|\beta\|^2$$

The ridge solution

$$\hat{\beta}_{ridge} = (X^T X + \lambda I)^{-1} X^T y$$

3.2.2.2 Lasso Regression

A ridge solution can be hard to interpret because it is not sparse (no β 's are set exactly to 0). What if we constrain the L1 norm instead of the Euclidean (L2) norm?

The objective is to minimize the sum of the squared errors

$$\min_{\beta} \|\varepsilon\|^2 = \min_{\beta} \|Y - X\beta\|^2, \text{ s. t. } \|\beta\|_1 \leq c$$

For $p=2$, the constraint in ridge regression corresponds to a circle, $\sum_{j=1}^p |\beta_j| \leq c$

We want to minimize:

$$\min_{\beta} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2, \text{ s. t. } |\beta_0| + |\beta_1| \leq c$$

Geometric Interpretation

The lasso performs L1 shrinkage, so that there are "corners" in the constraint, which in two dimensions corresponds to a diamond. If the sum of squares "hits" one of these corners, then the coefficient corresponding to the axis is shrunk to zero.

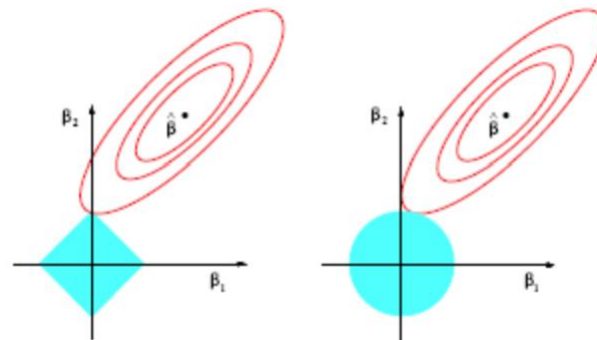


Figure 3.3 Estimation picture of the Lasso (left) and ridge regression (right). Shown are contours of the error and constraint functions. The solid blue areas are the constraint regions $|\beta_1| + |\beta_2| \leq c$ and $\beta_1^2 + \beta_2^2 \leq c^2$, respectively, while the red ellipses are the contours of the least squares error function. Lasso shrinks some coefficients to zero, resulting in sparse models.

As p increases, the multidimensional diamond has an increasing number of corners, and so it is highly likely that some coefficients will be set equal to zero. Hence, the lasso performs shrinkage and (effectively) subset selection.

Instead of solving the constrained form, we rewrite it using a Lagrange multiplier (λ):

$$\min_{\beta} \|Y - X\beta\|^2 + \lambda \|\beta\|_1$$

Unlike ridge regression, Lasso does not have a simple closed-form solution because the **L1-norm is not differentiable** at zero. Instead, numerical optimization techniques are used, like gradient descent.